

Transition Analysis Toolkit

A Reproducible Analysis of Student Transition Pathways

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1 Introduction

The transition from secondary school to first-year university mathematics remains a significant challenge for many students entering science, engineering, and related disciplines. Differences in prior mathematical preparation, curriculum pathways, and academic readiness contribute to wide variation in student success during the first year of university study. For institutions, understanding these transition pathways is important not only for improving student outcomes, but also for informing curriculum design, targeted support programs, and evidence-based educational practice. This study is motivated by the need to better understand the factors associated with successful transition into first-year university mathematics and to provide insights that can support both students and educators.

Over the past two decades, considerable research has examined factors associated with success in first-year university mathematics, including prior academic achievement, secondary school mathematics background, demographic characteristics, and institutional measures of academic preparedness. More recently, advances in educational data analytics and machine learning have enabled increasingly sophisticated predictive models of student performance. While these approaches have improved predictive capability, there remains a need for analyses that not only identify influential predictors but also interpret their relative importance in ways that are meaningful for educational decision-making. This study addresses that need through a transparent and reproducible analytical framework that combines predictive modelling with educational interpretation to better understand transition pathways into first-year university mathematics.

This paper investigates factors associated with successful transition into a first-stage undergraduate mathematics subject using a reproducible analytical workflow that integrates statistical analysis, machine learning, independent verification, and automated reporting. Rather than relying on a single predictive model, multiple tuned decision tree and random forest models are compared to evaluate both predictive performance and the consistency of feature importance across modelling approaches. By combining robust analytical methods with educational interpretation, the study aims not only to identify predictors of student success but also to provide practical insights that may inform curriculum design, transition support, and future research in first-year mathematics education.

1.1 Research Questions

1. Which factors are the strongest predictors of success in first-year university mathematics?
2. How consistent are predictor importance rankings across multiple machine learning models?
3. What educational insights can be drawn from the tuned models?

2 Methodology

This study adopted a reproducible analytical workflow to investigate factors associated with successful transition into first-year university mathematics. The workflow was designed to ensure that each stage of the analysis, from data preparation through statistical modelling, independent verification, and report generation, could be reproduced from the original research dataset using an automated build process. This approach emphasises transparency, repeatability, and traceability, allowing both analytical results and publication outputs to be generated consistently from the same underlying evidence.

2.1 Research Workflow

This study followed a reproducible analytical workflow consisting of data preparation, statistical analysis, machine learning model development, independent verification using R, and automated generation of publication-ready tables, figures, and manuscript components. Each stage formed part of an integrated build process, ensuring that analytical outputs could be reproduced directly from the underlying research dataset.

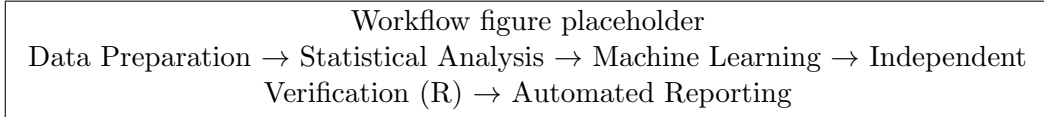


Figure 1: Overview of the reproducible analytical workflow used in this study.

2.2 Research Dataset

The study used de-identified institutional data from students enrolled in a first-stage undergraduate mathematics subject offered during the Autumn 2026 teaching session. The dataset was prepared specifically for research purposes using an approved de-identification process that removed all personally identifying information while preserving the academic and demographic characteristics required for analysis. The resulting dataset provided the foundation for all subsequent statistical analyses, machine learning models, independent verification procedures, and automated reporting undertaken in this study.

2.3 Variables

The analytical dataset comprised variables describing students' prior academic preparation, previous university mathematics experience, demographic characteristics, and current academic outcomes. Variables were selected to reflect factors commonly associated with transition into first-year university mathematics and were derived from institutional records following the approved research data preparation process. A complete description of each variable is provided in Table 1.

Table 1: Summary of variables included in the analytical dataset.

Variable	Description
<i>Placeholder for variable summary table</i>	

2.4 Statistical Analysis

Statistical analyses were undertaken to summarise the characteristics of the research dataset and to examine relationships between student background variables and academic outcomes. Descriptive statistics were generated to provide an overview of the cohort, while inferential analyses supported comparison of predictor variables prior to the development of machine learning models. These analyses established a statistical baseline against which the predictive models could subsequently be interpreted.

2.5 Machine Learning Models

Machine learning models were developed to investigate the relative importance of factors associated with successful transition into first-year university mathematics. Decision tree and random forest classifiers were selected because they provide both competitive predictive performance and interpretable measures of feature importance. Hyperparameter tuning and model

evaluation were undertaken using reproducible workflows to ensure consistent comparison across modelling approaches.

2.6 Independent Verification

To strengthen confidence in the analytical results, an independent verification pipeline was implemented using the R statistical computing environment. Key statistical summaries and model outputs were reproduced independently of the primary Python implementation to confirm the consistency and reliability of the analytical workflow. This independent verification formed an additional quality assurance stage prior to the generation of publication outputs.

2.7 Automated Reporting

The final stage of the analytical workflow automatically generated publication-ready tables, figures, model summaries, and manuscript components directly from the verified analytical outputs. By integrating analysis and reporting within a single reproducible workflow, the potential for manual transcription errors was reduced while ensuring that all reported results remained synchronised with the underlying research data and analyses.

The methodology described above established a transparent and reproducible analytical framework for investigating factors associated with successful transition into first-year university mathematics. The following section presents the outcomes of this workflow, beginning with an evaluation of predictive model performance before examining the relative importance and educational interpretation of the identified predictors.

3 Results

This section presents the findings generated by the reproducible analytical workflow described in the preceding methodology. Results are presented in four stages. First, the predictive performance of the machine learning models is evaluated. Second, the relative importance of predictor variables is examined. Third, the consistency of feature rankings across modelling approaches is assessed. Finally, the educational significance of these findings is interpreted in the context of student transition into first-year university mathematics.

3.1 Model Performance

Table 2: ROC AUC comparison for cross-validated Decision Tree and Random Forest models.

Model	CV ROC AUC	Test ROC AUC	Test Rows
Decision Tree — With Risk	0.7072	0.7664	295
Random Forest — Without Risk	0.7101	0.7603	295
Random Forest — With Risk	0.7095	0.7599	295
Decision Tree — Without Risk	0.6994	0.7557	295

3.2 Feature Importance

Rank	Research theme	Mean importance
1	Secondary mathematics preparation	0.2755
2	Composite risk measure	0.2381
3	Preparatory mathematics	0.2320
4	Previous university performance	0.1090
5	Secondary school achievement	0.1058
6	Student characteristics	0.0395

3.3 Feature Rankings

Rank	Feature	Models	Mean importance
1	Composite risk index	2	0.4762
2	Prior preparatory mathematics	4	0.1694
3	Secondary mathematics: Other	4	0.1019
4	Secondary mathematics: Standard	4	0.0950
5	Previous attempt at current subject	4	0.0693
6	Previous preparatory mathematics pass	4	0.0532
7	Previous failure in current subject	4	0.0397
8	International student	4	0.0395
9	Secondary school mark below 70	4	0.0394
10	Secondary mathematics: Extension 1	4	0.0348
11	Secondary school mark 90 or above	4	0.0288
12	Secondary mathematics: Extension 2	4	0.0257
13	Secondary school mark 80–89	4	0.0228
14	Secondary mathematics: Advanced	4	0.0182
15	Secondary school mark 70–79	4	0.0148
16	Previous preparatory mathematics failure	4	0.0094

3.4 Model Interpretation

Across the tuned tree-based models, secondary mathematics preparation emerged as the most influential predictor group (mean importance = 0.276). This was followed by the composite risk measure (0.238) and preparatory mathematics (0.232). Together, these three groups accounted for 74.6% of the average feature importance across the tuned models. Secondary mathematics preparation remained influential across all four models, whereas the composite risk measure contributed strongly but appeared in two of the four models. In contrast, student characteristics made the smallest average contribution. This suggests that prior secondary mathematics preparation was the strongest and most consistent predictor of success across the tuned models.

4 Discussion

The results of this study provide new insights into the factors associated with successful transition into first-stage university mathematics. By combining reproducible analytical methods with interpretable machine learning models, the analysis extends beyond prediction alone to examine the relative importance and consistency of the variables associated with student success. This discussion considers these findings in the context of previous research, reflects on their educational significance, and identifies implications for both institutional practice and future research.

As demonstrated in Section 3.2, secondary mathematics preparation emerged as the strongest and most influential predictor group across the tuned models, with a mean feature importance of 0.276. Composite risk (0.238) and preparatory mathematics (0.232) were the next most influential predictor groups. Collectively, these three groups accounted for 74.6% of the average feature importance, indicating that measures of prior mathematical preparation dominated the predictive performance of the evaluated models. Furthermore, the consistency of the feature rankings presented in Section 3.3 showed that secondary mathematics preparation remained influential across all four tuned models, whereas the composite risk measure contributed strongly but appeared in only two models. Together, these findings provide robust evidence that students' secondary mathematical preparation was the strongest and most consistent predictor of success in this cohort.

5 Conclusion

References

A Supplementary Material